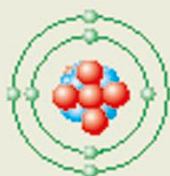


Chemistry

All matter is made of tiny particles called atoms. Chemistry is the branch of science that studies the structure and properties of atoms. It also investigates how atoms change and interact with each other during processes called chemical reactions.

Types of chemistry

There are several different branches of chemistry that investigate the composition, behavior, and properties of different types of matter. Three of the main areas of study are described below.



Organic chemistry

This branch of chemistry examines all the compounds that contain carbon. Most carbon compounds are derived from organic sources (meaning it comes from living things).



Electrochemistry

Electrochemists examine the relationship between electricity and chemical substances, and investigate the chemical processes that cause electrons to move.

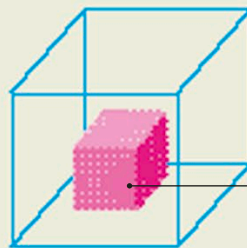


Inorganic chemistry

Most matter in the Universe is nonliving—it does not grow, reproduce, or move. Inorganic chemistry is the study of all nonorganic substances (meaning they come from nonliving things).

States of matter

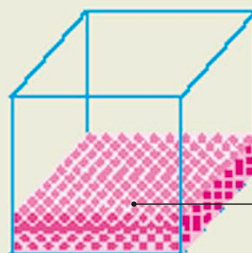
There are three main states of matter—solid, liquid, and gas. The state of a substance depends on how its atoms and molecules (groups of atoms) are bonded together. This bonding is determined by factors such as temperature and pressure.



Solid

A solid has a fixed shape or volume. Its molecules are tightly packed together in regular patterns. Strong bonds hold the solid firmly in place.

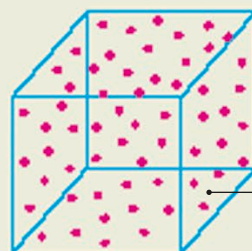
Molecules are compacted together



Liquid

A liquid has a fixed volume but no fixed shape—it takes the shape of its container. The bonds of a liquid are looser than those of a solid.

Some bonds are broken, which gives liquid its fixed volume, but fluid shape.



Gas

A gas has no fixed volume or shape. There are no bonds holding the molecules together, so they move freely around their container.

Without bonds, the molecules are free to move in any direction.

Chemical properties

All substances have their own unique set of chemical properties, which help to explain why they react in a particular way when they are heated or combined with other substances.

Reactivity series

Common metals are often ordered by how reactive they are. This is called the reactivity series. The metals at the top of the series are the most reactive; those at the bottom are the least reactive.

